

2012 HKDSE Physics

Paper 2 Elective Sections

Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

Section A Astronomy and Space Science

Question 1.1

Weightlessness occurs inside a spacecraft orbiting the Earth. Which statement is correct?

- A. Weightlessness only occurs for objects inside a spacecraft orbiting the Earth.
- B. The Earth's gravitational attraction in the orbit is so weak that the gravitational force is practically zero.
- C. The Earth's gravitational attraction is cancelled by that of the Moon.
- D. Both the spacecraft and the objects inside undergo free fall towards the Earth.

Official answer: D

D. Gravity supplies the centripetal acceleration. The spacecraft and its contents fall together, so no supporting normal force is needed; their apparent weight is zero. Gravity itself is not zero.

Question 1.2

An interplanetary spacecraft is launched from Earth with initial speed $\sqrt{3GM/R}$, where G is the gravitational constant, M the Earth's mass and R its radius. What is its speed very far from Earth?

- A. $\sqrt{GM/R}$
- B. $\sqrt{GM/(2R)}$
- C. $\sqrt{2GM/R}$
- D. Zero

Official answer: A

A. Conservation of mechanical energy gives $\frac{1}{2}m(3GM/R) - GMm/R = \frac{1}{2}mv_{\infty}^2$. Hence $v_{\infty}^2 = GM/R$ and $v_{\infty} = \sqrt{GM/R}$.

Question 1.3

The Sun is about 8 kpc from the Milky Way's centre and moves around it at 220 km s^{-1} . How long does one revolution take?

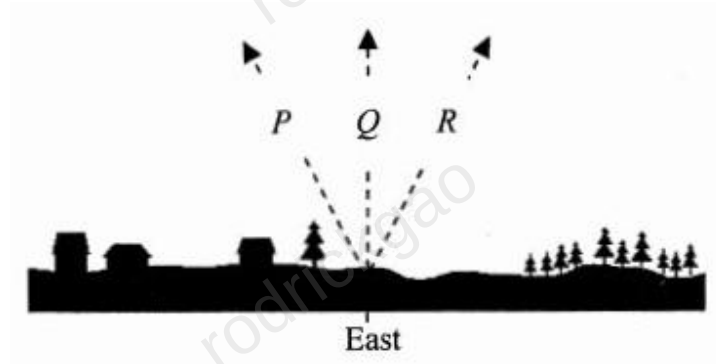
- A. 2.24×10^8 years
- B. 3.55×10^8 years
- C. 2.24×10^{11} years
- D. 3.55×10^{11} years

Official answer: A

A. $T = 2\pi r/v = 2\pi(8 \times 10^3 \times 3.09 \times 10^{16})/(220 \times 10^3) = 7.06 \times 10^{15} \text{ s} \approx 2.24 \times 10^8 \text{ years}$.

Question 1.4

The figure shows the horizon when facing east in Hong Kong. Which arrow represents the direction in which stars rise?



- A. P
- B. Q
- C. R
- D. The direction varies with the seasons.

Official answer: C

C. Hong Kong is north of the equator. When looking east, north is to the left and south to the right. The daily stellar paths rise obliquely towards the south, so the rising direction is R.

Question 1.5

Which statement about the Earth's motion around the Sun is INCORRECT?

- A. The orbital speed is not constant.
- B. The Sun is at the centre of the Earth's orbit.
- C. The Earth–Sun distance changes periodically.
- D. In general the Earth's instantaneous velocity is not perpendicular to the Sun's gravitational force.

Official answer: B

B. The orbit is an ellipse with the Sun at one focus, not at its centre. Conservation of angular momentum gives a greater speed near perihelion. In general the velocity has a radial component; it is perpendicular to the gravitational force only at the apsides.

Question 1.6

Which statements about stars X and Y are correct?

Star	Absolute magnitude	Apparent magnitude
X	2.8	4.7
Y	5.4	3.2

- (1) X is farther from Earth than Y.
- (2) Y is farther from Earth than X.
- (3) Their distances from Earth can be determined from the information.

- A. (1) only
 B. (2) only
 C. (1) and (3) only
 D. (2) and (3) only

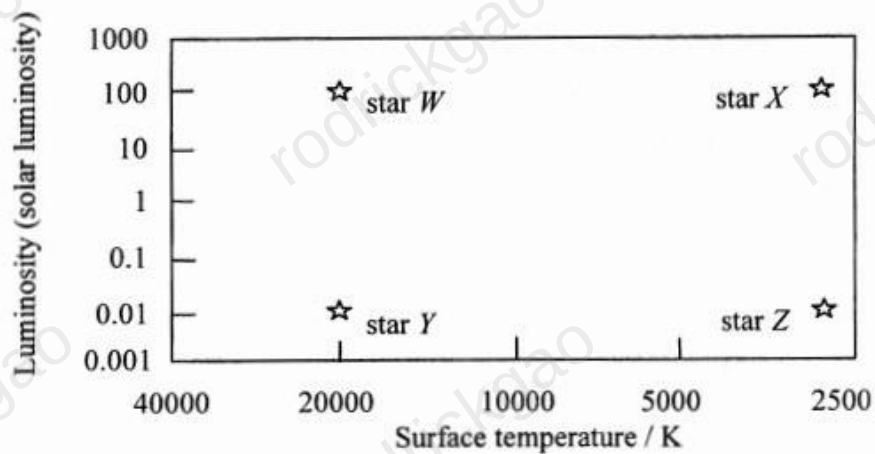
Official answer: C

C. $m - M = 5 \log_{10}(d/10 \text{ pc})$, so $d = 10 \times 10^{(m-M)/5} \text{ pc}$. Thus $d_X \approx 24.0 \text{ pc}$ and $d_Y \approx 3.63 \text{ pc}$. Both distances are calculable and X is farther away.

Question 1.7

The diagram gives the luminosities and surface temperatures of stars W, X, Y and Z. Which statements are correct?

- (1) For X, red light has higher intensity than any other colour.
- (2) For W, blue light has higher intensity than any other colour.
- (3) The intensity ratio of red light to other colours is larger for Z than for Y.



- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)

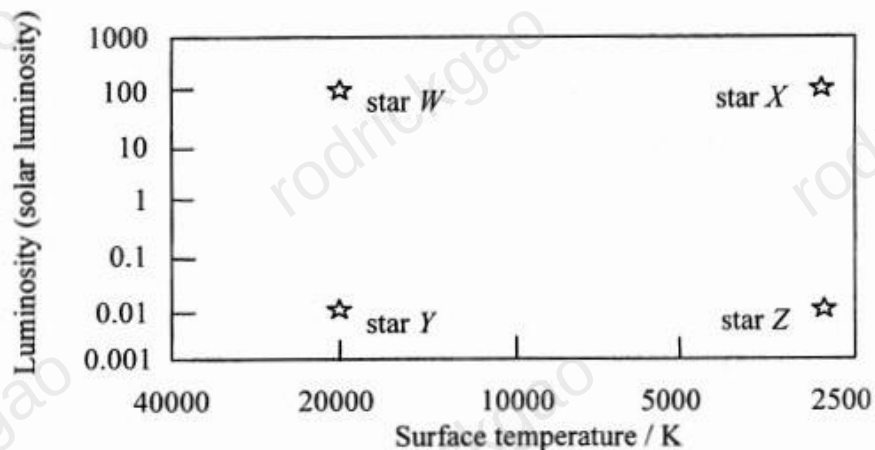
Official answer: D

D. Colour is governed by surface temperature. Cool X and Z (about 3000 K) favour red within the visible band, whereas hot W and Y (about 20 000 K) favour blue. Thus all three statements hold. This comparison concerns visible colours; the overall blackbody peak need not be visible.

Question 1.8

The diagram gives information about stars W, X, Y and Z. The absorption spectrum of X contains hydrogen absorption lines. What can be concluded?

- (1) X is mainly composed of hydrogen.
- (2) Hydrogen gas is present in X's outer atmosphere.
- (3) X has a lower hydrogen abundance than a similar star of the same temperature.



- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: B

B. Hydrogen in the cooler outer atmosphere absorbs particular wavelengths from the continuous spectrum passing through it. The presence of lines identifies hydrogen, but alone does not establish the bulk composition or a relative abundance.

Question 1 Structured Question

(a) Let R_s , T_s , L_s be the Sun's radius, surface temperature and luminosity, and R , T , L those of another star.

(i) Show that $R = (T_s/T)^2(L/L_s)^{1/2}R_s$. (2 marks)

(ii) Betelgeuse has surface temperature 3650 K and luminosity 126000 L_s . Find its radius in terms of R_s , taking $T_s = 5780$ K. (2 marks)

(b)(i) The luminosity above corresponds to an estimated distance of 197 pc. A 2008 measurement was 197 ± 45 pc. Without calculating the actual radius, explain how the radius in (a)(ii) changes if the upper distance limit is used. Treat Betelgeuse as a point source emitting evenly in all directions. (2 marks)

(ii) Suggest why its distance is difficult to measure accurately by parallax. (1 mark)

(c) In 2011, media reports suggested that a Betelgeuse supernova would appear as a “second sun” for a few weeks. Assess this by comparing its brightness with the Sun's. A similar-mass star has supernova luminosity about $10^9 L_s$ for a certain time, with about 1% of the explosion power appearing as visible light. Take Betelgeuse's distance as 200 pc. (3 marks)

Official marking scheme

Marks

1. (a) (i) $L_S = \sigma T_S^4 (4\pi R_S^2)$

1M

$$L = \sigma T^4 (4\pi R^2)$$

$$\therefore \frac{L_S}{L} = \frac{T_S^4 R_S^2}{T^4 R^2}$$

1M

$$R = \left(\frac{T_S}{T} \right)^2 \left(\frac{L}{L_S} \right)^{\frac{1}{2}} R_S$$

2

(ii)

Sun	$T_S = 5780 \text{ K}$	L_S	R_S
Betelgeuse	$T = 3650 \text{ K}$	$L = 126000 L_S$	R

$$R = \left(\frac{5780}{3650} \right)^2 \left(\frac{126000 L_S}{L_S} \right)^{\frac{1}{2}} R_S$$

$$= 890 R_S$$

1M

1A 2

- (b) (i) $\therefore$ Same brightness measured and brightness $= \frac{L}{4\pi d^2}$ or $\propto \frac{L}{d^2}$ or L increases with d ,
 d is larger $\rightarrow L$ is larger. (for d taken as $(197+45) \text{ pc}$)

1M

i.e. for Betelgeuse, $\therefore L \propto R^2$ $\therefore R$ increases.

1A 2

Or

$$M = m - 5(\log_{10} d - 1) \quad \text{or} \quad M = m - 5(\log_{10} \frac{d}{10})$$

(M : absolute magnitude, m : apparent magnitude)

- (ii) The parallax measurement ($d = \frac{1}{p}$, accurate to within $\sim 100 \text{ pc}$) is too small or d is too large (too far away), ($\sim (1/200)'' = 5 \text{ milliarcsec}$).

1A 1

- (c) $L = 10^9 L_S$, brightness $= \frac{(0.01 \times 10^9 L_S)}{4\pi d^2}$
- $$d = 200 \times 206265 \text{ AU} = 41253000 \text{ AU}$$
- $$\text{Brightness} = \frac{(0.01 \times 10^9)}{41253000^2} \frac{L_S}{4\pi (1\text{AU})^2} = \frac{(0.01 \times 10^9)}{41253000^2} \text{ brightness of the Sun}$$
- $$= 5.88 \times 10^{-9} \text{ brightness of the Sun}$$

1M

1M

1A

3

(a)(i) Apply the Stefan–Boltzmann law to each spherical star and divide to cancel σ and 4π : $L/L_S = (R/R_S)^2(T/T_S)^4$. Taking the positive square root gives the required radius.

(a)(ii) $R/R_S = (5780/3650)^2 \sqrt{126000} = 890.15 \approx 890$. The much lower temperature requires a very large area for this luminosity.

(b)(i) At fixed measured flux and temperature, $L \propto d^2$ and $R \propto \sqrt{L} \propto d$. Therefore using the upper distance limit increases the inferred radius.

(b)(ii) At about 200 pc the parallax is only about 0.005 arcsec. A given angular uncertainty therefore produces a large fractional uncertainty in distance. The quoted 100 pc range is the historical scheme's context, not a universal limit for all instruments.

(c) Compare received fluxes, not luminosities. The distance in AU is 4.1253×10^7 ; the scheme's visible-supernova/Sun brightness ratio is $10^7/(4.1253 \times 10^7)^2 = 5.876 \times 10^{-9}$. It is far fainter than the Sun, so the “second sun” claim is false under this comparison. The scheme uses total solar flux as its reference.

Source discrepancy, (c): the Chinese marking scan prints 5.88×10^9 in its final line, missing the minus sign. The English marking scan and the preceding calculation give 5.88×10^{-9} . The teaching answer uses the latter.

Section B Atomic World

Question 2.1

From the classical viewpoint, what are the limitations of Rutherford's atomic model?

- (1) Atoms would continuously emit electromagnetic radiation.
- (2) Atoms would be unstable and eventually collapse.
- (3) The atomic emission spectrum would be continuous rather than discrete.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: D

D. An orbiting electron accelerates and, classically, radiates continuously. As it loses energy it spirals inward, making the atom unstable. The continuously changing orbital frequency would yield a continuous spectrum, contrary to observed line spectra.

Question 2.2

Which statements about spectra are correct?

- (1) A tungsten-filament lamp emits a continuous spectrum.
- (2) Viewing a tungsten-filament lamp through hydrogen gas can give a line absorption spectrum.
- (3) Hydrogen's emission spectrum consists of dark lines on a bright background.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

A. A hot dense filament emits a continuous thermal spectrum. Cooler hydrogen in front absorbs selected photon energies and creates dark absorption lines. Hydrogen emission instead consists of bright lines on a dark background, so (3) is false.

Question 2.3

Photons of energy 7 eV strike a photocell cathode and emit photoelectrons with maximum kinetic energy 4 eV. What is the stopping potential when 4 eV photons strike the same cathode?

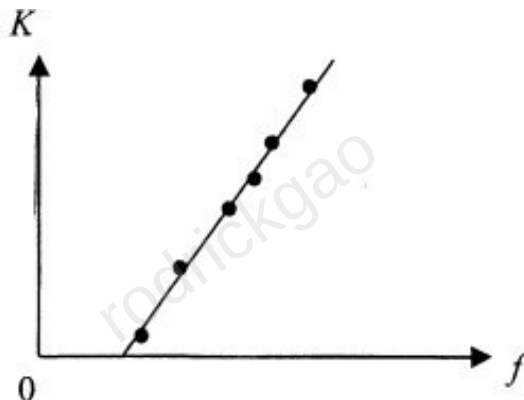
- A. 0 V
- B. 1 V
- C. 2 V
- D. 3 V

Official answer: B

B. The work function is $\phi = 7 - 4 = 3$ eV. With 4 eV photons, $K_{\max} = 4 - 3 = 1$ eV. Since $eV_s = K_{\max}$, the stopping-potential magnitude is 1 V.

Question 2.4

Ultraviolet radiation of different frequencies f illuminates a metal. The graph shows the maximum photoelectron kinetic energy K against f . How does using higher-intensity ultraviolet radiation on the same metal affect the slope and horizontal-axis intercept?



- A. Slope smaller; intercept unchanged.
- B. Slope larger; intercept unchanged.
- C. Slope unchanged; intercept unchanged.
- D. Slope unchanged; intercept smaller.

Official answer: C

C. $K_{\max} = hf - \phi$. The slope h and threshold frequency ϕ/h depend on fundamental constants and the metal, not light intensity. Higher intensity increases photon arrival rate and hence photocurrent, not the maximum energy at a given frequency.

Question 2.5

According to Bohr's hydrogen model, what is the ratio of the electron-orbit radius in the first excited state to that in the second excited state?

- A. 1 : 2
- B. 1 : $\sqrt{2}$
- C. 4 : 9
- D. 2 : 3

Official answer: C

C. The first and second excited states have $n = 2$ and $n = 3$ respectively, not $n = 1$ and 2. Since $r_n \propto n^2$, $r_2 : r_3 = 4 : 9$.

Question 2.6

Which has the shortest de Broglie wavelength?

- A. A 60 kg person walking at 0.8 m s^{-1} .
- B. A 0.3 kg bird flying at 20 m s^{-1} .
- C. A 0.6 kg basketball moving at 12 m s^{-1} .
- D. A 0.05 kg bullet moving at 800 m s^{-1} .

Official answer: A

A. $\lambda = h/p = h/(mv)$. Momenta for A–D are 48, 6, 7.2 and 40 kg m s^{-1} . A has the largest momentum and thus the shortest wavelength, approximately $1.38 \times 10^{-35} \text{ m}$.

Question 2.7

Which property explains the Lotus effect?

- A. Water-attractive property
- B. Water-repelling property**
- C. Wave-particle duality of matter
- D. High electrical conductivity

Official answer: B

B. The hydrophobic micro/nanostructured surface allows water to form droplets that roll off, carrying dirt away. This water-repelling behaviour produces the self-cleaning Lotus effect.

Question 2.8

A substance is reduced to particles about 10 nm in size. Which properties can differ from those of the bulk substance?

- (1) Optical
 - (2) Mechanical
 - (3) Electrical
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)**

Official answer: D

D. At nanometre scales the fraction of surface atoms rises strongly and electronic behaviour can change through confinement. These effects can alter optical absorption, mechanical response and electrical conduction, so all three categories can differ from the bulk.

Question 2 Structured Question

The electron energy levels in hydrogen are $E = -13.6/n^2$ eV.

- (a) Explain the physical meaning of the negative sign of E . (1 mark)
- (b) State TWO non-classical postulates of Bohr's hydrogen model. (2 marks)
- (c) Ground-state hydrogen is illuminated with ultraviolet light of wavelengths 102.8 nm and 100.0 nm. The 102.8 nm light is absorbed but the 100.0 nm light is unaffected.
 - (i) Calculate the energy in eV of a 102.8 nm photon and determine the atom's quantum number after absorption. (3 marks)
 - (ii) Why does 100.0 nm light pass through without absorption? (1 mark)
 - (iii) As the excited atom returns to the ground state, how many transitions are possible? Identify and explain which gives visible light. Visible-photon energies range from 1.7 eV to 3.2 eV. (3 marks)

Official marking scheme

Marks

2. (a) The negative sign means that:
- the electron is "bound" to the atom
 - the force between the nucleus and electron is attractive
 - work has to be done to remove the electron to infinity
- } ANY ONE 1A 1
- (b)
- The angular momenta of the electrons are whole number multiples of $\frac{h}{2\pi}$, i.e. quantized.
 - The electrons are in stable orbits without emitting radiation.
 - The electrons can occupy certain discrete orbits only. / Total energy of the atom is quantized / Energy levels are discrete.
 - The atom can only emit or absorb radiation in the form of a photon when an electron jumps from one energy level to another.
- } ANY TWO 2A 2
- (c) (i) Energy = hf
- $$= \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{102.8 \times 10^{-9}} = 1.93 \times 10^{-18} \text{ J}$$
- $$= 1.93 \times 10^{-18} / (1.60 \times 10^{-19}) = 12.09 \text{ (eV)}$$
- $$\Delta E = 12.09 \text{ eV} = -\left(\frac{1}{n^2} - \frac{1}{1^2}\right) 13.6 \text{ eV}$$
- $$n^2 = \frac{1}{1 - \frac{12.09}{13.6}}$$
- $$n^2 = 9.007 \Rightarrow n = 3$$
- 1A 3
- (ii) The energy of UV = 100.0 nm does not match the differences between ground state of hydrogen and other energy levels. 1A 1
- (iii) There are three possible transitions 1A
- The hydrogen atom is in excited state $n = 3$.
- $$E_3 = -\frac{13.6}{3^2} \text{ eV} = -1.51 \text{ eV}$$
- $$E_2 = -\frac{13.6}{2^2} \text{ eV} = -3.40 \text{ eV}$$
- $$E_1 = -\frac{13.6}{1^2} \text{ eV} = -13.60 \text{ eV}$$
- $3 \rightarrow 1 \quad \Delta E = 12.09 \text{ eV} (= -1.51 - (-13.6))$
 $3 \rightarrow 2 \quad \Delta E = 1.89 \text{ eV} (= -1.51 - (-3.40))$
 $2 \rightarrow 1 \quad \Delta E = 10.2 \text{ eV} (= -3.40 - (-13.6))$
- 1M
- The transition from $3 \rightarrow 2$ corresponds to visible light as 1.89 eV is within the corresponding range. 1A 3

- (a) Zero energy corresponds to a stationary electron infinitely far from the nucleus. A negative total energy means it is bound and energy must be supplied to ionize the atom.
- (b) Two examples are stable non-radiating stationary states and quantized angular momentum $mvr = nh/(2\pi)$. In contrast, classical accelerating charges radiate and classical orbital angular momentum is unrestricted.
- (c)(i) Using the supplied constants, $hc/(\lambda e) = 12.092 \text{ eV}$. The final energy is $-13.6 + 12.092 \approx -1.508 \text{ eV}$, matching $E_3 = -1.511 \text{ eV}$ within rounding, hence $n = 3$.
- (c)(ii) A 100.0 nm photon has 12.431 eV . This lies between the $1 \rightarrow 3$ gap (12.089 eV) and $1 \rightarrow 4$ gap (12.75 eV), and below the 13.6 eV ionization threshold. It cannot be partly absorbed to fit a discrete transition.
- (c)(iii) The available downward transitions among $n = 3, 2, 1$ are $3 \rightarrow 1$, $3 \rightarrow 2$ and $2 \rightarrow 1$. Only $13.6(1/2^2 - 1/3^2) = 1.889 \text{ eV}$ lies in $1.7\text{--}3.2 \text{ eV}$, so only $3 \rightarrow 2$ is visible. A single atom follows either the direct path or a cascade; it does not emit all three photons in one decay.

Section C Energy and Use of Energy

Question 3.1

Which lamp has the greatest end-use energy efficiency? Options give luminous flux and power rating.

A. 750 lm; 15 W

B. 900 lm; 30 W

C. 750 lm; 60 W

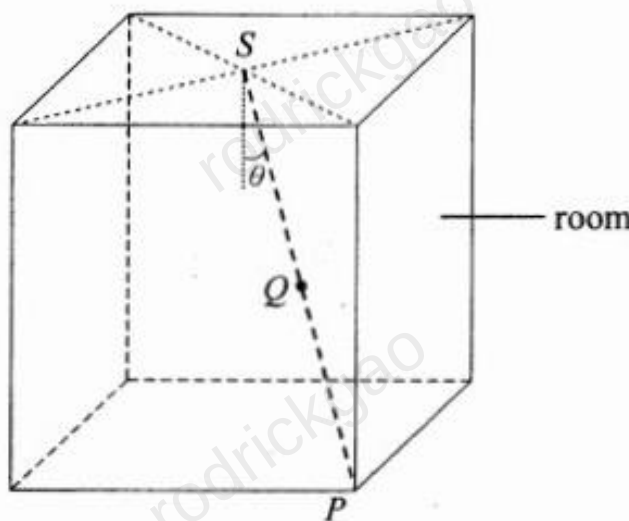
D. 600 lm; 90 W

Official answer: A

A. Compare luminous efficacy Φ/P : $750/15 = 50$, $900/30 = 30$, $750/60 = 12.5$ and $600/90 = 6.67 \text{ lm W}^{-1}$. A delivers the most visible-light output per watt.

Question 3.2

A point source S illuminates a room. The illuminance at corner P is E. Neglect reflections from room surfaces. Q is midway between P and S. What is the illuminance at Q?



A. $\sqrt{2} E / \cos \theta$

B. $2E \cos \theta$

C. $4E$

D. $8E$

Official answer: C

C. For the same receiving-surface orientation, illuminance is $I \cos \theta / r^2$. P and Q lie on the same ray so θ is unchanged, while $SQ = SP/2$. Hence $E_Q/E_P = (SP/SQ)^2 = 4$.

Question 3.3

Which sequence best describes energy conversion during regenerative braking in an electric vehicle?

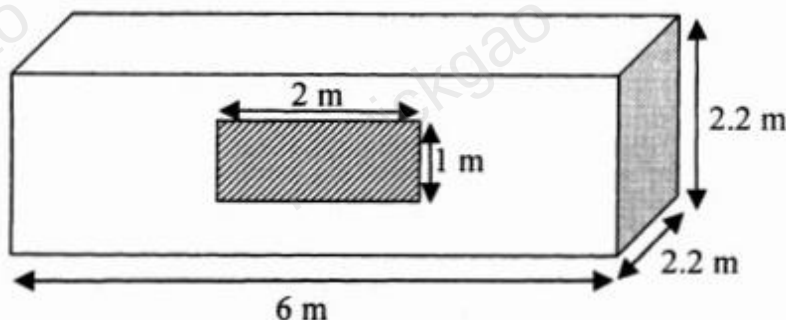
- A. Kinetic $\rightarrow$ electrical
- B. Kinetic $\rightarrow$ chemical
- C. Kinetic $\rightarrow$ chemical $\rightarrow$ electrical
- D. Kinetic $\rightarrow$ electrical $\rightarrow$ chemical

Official answer: D

D. During braking the motor acts as a generator, converting the vehicle's kinetic energy into electrical energy. Charging the battery then stores this energy chemically.

Question 3.4

A $6\text{ m} \times 2.2\text{ m} \times 2.2\text{ m}$ container is converted into an office. A $1\text{ m} \times 2\text{ m}$ glass window is installed in the front. The equivalent temperature difference across each of the five exposed surfaces is 7°C . Ignore solar radiation entering through the window. The U-values of the metal and glass are 26.2 and $1.8\text{ W m}^{-2}\text{ K}^{-1}$ respectively. Estimate the office's overall thermal transfer value (OTTV), in W m^{-2} .



- A. 25.2
- B. 26.2
- C. 176.5
- D. 183.4

Official answer: C

C. The five exposed faces comprise four walls and the roof: $A = 2(6 \times 2.2) + 2(2.2 \times 2.2) + 6 \times 2.2 = 49.28\text{ m}^2$. Glass area is 2 m^2 and metal area 47.28 m^2 . OTTV = $[26.2(47.28) + 1.8(2)] \times 7 / 49.28 = 176.47 \approx 176.5\text{ W m}^{-2}$. The floor is excluded.

Question 3.5

Estimate the time to cool the air in an insulated room of interior volume 29.0 m^3 from 37°C to 24°C using an air conditioner of cooling capacity 2.2 kW . Air density is 1.2 kg m^{-3} and specific heat capacity is $1000\text{ J kg}^{-1}\text{ K}^{-1}$.

- A. 171 s
- B. 206 s
- C. 380 s
- D. 586 s

Official answer: B

B. The air mass is $\rho V = 1.2 \times 29.0 = 34.8\text{ kg}$. Removing $Q = mc\Delta T = 34.8 \times 1000 \times 13 = 452400\text{ J}$ at 2200 W takes $t = Q/P = 205.6 \approx 206\text{ s}$. This estimate considers the air alone.

Question 3.6

A wind turbine has blades 5 m long. Wind blows normally at 12 m s^{-1} ; the overall efficiency is 25%. Estimate how many turbines are needed for 1 MW electrical output. Air density is 1.2 kg m^{-3} .

A. 12

B. 49

C. 122

D. 196

Official answer: B

B. Swept area $A = \pi(5)^2 \text{ m}^2$. Each turbine supplies $\eta(\frac{1}{2}\rho A v^3) = 0.25 \times \frac{1}{2} \times 1.2 \times \pi \times 25 \times 12^3 \approx 20358 \text{ W}$. Thus $N = 10^6/20358 \approx 49.1$, matching the requested estimate 49.

Question 3.7

A hydroelectric station has a water-level difference of 50 m. Water passes through the turbine at the lower level at 3000 kg s^{-1} . Estimate the turbine efficiency if output power is 1 MW. Take $g = 9.81 \text{ m s}^{-2}$.

A. 32%

B. 60%

C. 68%

D. 75%

Official answer: C

C. Gravitational power input $= mgh = 3000 \times 9.81 \times 50 = 1.4715 \times 10^6 \text{ W}$. $\eta = P_{\text{out}}/P_{\text{in}} = 10^6/(1.4715 \times 10^6) = 0.6796 \approx 68\%$.

Question 3.8

What would happen if the moderator of a nuclear fission reactor failed to function?

A. The chain reaction might eventually stop.

B. Neutrons could not be absorbed by the moderator.

C. Heat could not be transferred to the steam generator.

D. The fuel rods might melt down.

Official answer: A

A. The moderator slows fast neutrons to thermal energies, at which they are more likely to cause further fission in the fuel of a thermal reactor. If moderation fails, too few suitable neutrons remain to sustain the chain reaction. Neutron absorption is chiefly the control rods' role; heat transport is the coolant's role.

Question 3 Structured Question

- (a) Even on a clear day the atmosphere absorbs an average of 26.8% of solar power. Find the maximum solar power per unit area reaching Earth's surface. Solar constant = 1366 W m^{-2} . (1 mark)
- (b) State the energy conversion in a solar cell and suggest one way to improve its absorption of energy. (2 marks)
- (c) Solar Impulse is a Swiss project for a long-distance solar aircraft. Prototype HB-SIA has four battery-driven engines, with onboard solar cells charging the batteries. It made a successful international flight in May 2011. Each engine is rated 7.35 kW; each cell panel has area 0.0172 m^2 ; cell conversion efficiency is 12% at midday with normally incident solar radiation.
- (i) Assume all cell electrical output is shared equally by the four engines and all cells receive the power per unit area in (a). Estimate the number of cells needed to run all four engines at full power. (3 marks)
- (ii) Only 11628 cells were installed for the 2011 flight, insufficient to run all engines at full power. Suggest a practical reason for this design. (1 mark)
- (d) Explain why solar power is renewable. Apart from solar power, suggest the most feasible renewable energy source for Hong Kong and justify your choice. (3 marks)

Official marking scheme

	Marks
3. (a) Maximum solar power per unit area reaching the Earth $= 1366 \times (1 - 0.268)$ $= 1000 \text{ W m}^{-2}$ or W	1A 1
(b) Solar energy / radiation / radiant energy / light energy $\rightarrow$ electrical energy. Adhere a transparent anti-reflection film to the solar cell panel. Methods involving tracking the sun to receive maximum sun light Mirrors/lens to reflect/collect sun light to the solar cells	1A } ANY ONE 1A 2
(c) (i) Solar power received by each solar cell $= 1000 \text{ W m}^{-2} \times 0.0172 \text{ m}^2$ $= 17.2 \text{ W}$	1M
Electrical power delivered by each solar cell $= 17.2 \text{ W} \times 0.12$ $= 2.064 \text{ W}$	1M
Number of solar cells required $= \frac{7.35 \text{ kW} \times 4}{2.064 \text{ W}}$ $= 14244$	1A 3
(ii) - To limit/minimize the weight of the aircraft. - Limited area for installing the solar cells. - It is the batteries to deliver maximum power to the engines, the solar cells are for charging the batteries.	} ANY ONE 1A 1
(d) Energy comes from natural resources / processes that are replenished constantly.	1A
Wind power	1A
as (northeast and southwest) monsoons prevail in Hong Kong (during winter and summer).	1A 3

(a) The transmitted fraction is $1 - 0.268 = 0.732$. Thus intensity = $1366 \times 0.732 = 999.912 \approx 1000 \text{ W m}^{-2}$. Power per area must carry W m^{-2} .

(b) Photovoltaic cells convert light directly to electricity. An anti-reflection coating reduces the reflected fraction so more incident radiation enters the cell; tracking maintains a large projected collecting area.

(c)(i) Total engine demand = $4 \times 7350 = 29400 \text{ W}$. With the scheme's rounded intensity, each cell gives $1000 \times 0.0172 \times 0.12 = 2.064 \text{ W}$, so the estimate is $29400/2.064 = 14244.19$, reported officially as 14244. The question asks for an estimate. A strict integer minimum would be 14245 using 1000 W m^{-2} , or 14246 if 999.912 W m^{-2} is retained throughout.

(c)(ii) More cells add weight and require more area. The batteries can supply short periods of high demand, so the installed solar array need not equal the simultaneous maximum engine rating.

(d) Solar radiation is naturally and continuously supplied on human timescales. The scheme's alternative is wind power: Hong Kong's seasonal monsoons provide a recurring wind resource. This answers the historical examination prompt rather than comparing present-day projects.

Section D Medical Physics

Question 4.1

Joan cannot see objects clearly closer than 0.80 m from her eyes. What lens type and power should she wear to correct her near point to 0.25 m?

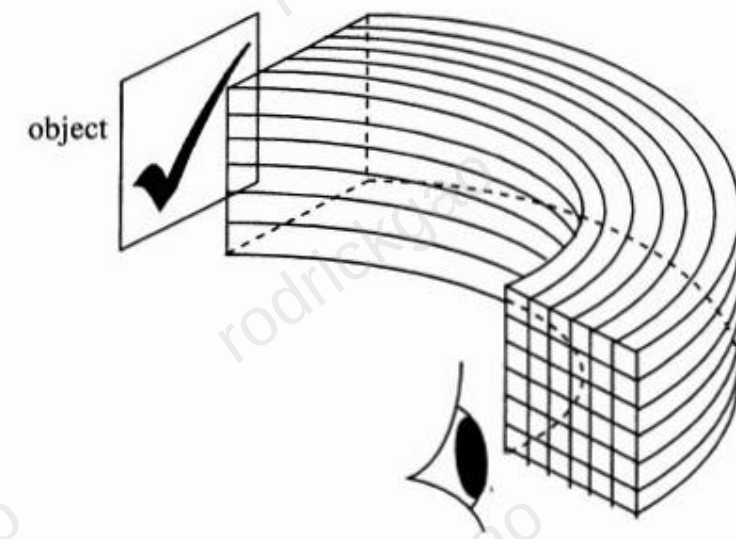
- A. Convex lens, +2.75 D
- B. Convex lens, +5.25 D
- C. Concave lens, -2.75 D
- D. Concave lens, -5.25 D

Official answer: A

A. An object at 0.25 m must form a virtual image at the unaided near point 0.80 m away. With $u = 0.25$ m and $v = -0.80$ m, $P = 1/f = 1/u + 1/v = 4 - 1.25 = +2.75$ D. Positive power means a convex lens.

Question 4.2

A coherent optical-fibre bundle contains 36 square elements and is used to view the object shown (not to scale). Which illustrated option best represents the picture seen by the observer?



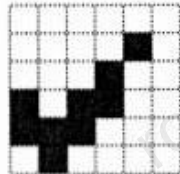
A.



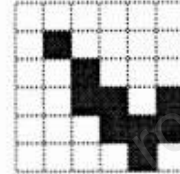
B.



C.



D.



A. Smooth tick, original orientation (diagram A).

B. Smooth tick, laterally reversed (diagram B).

C. Pixelated tick, original orientation (diagram C).

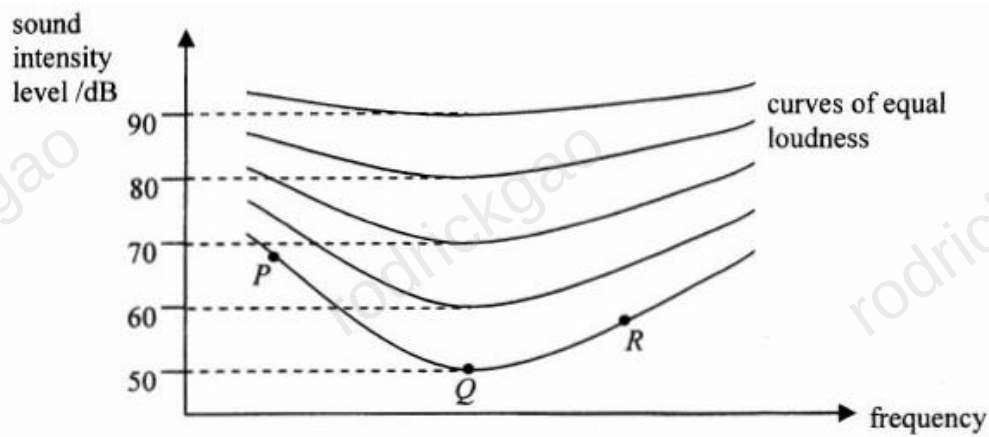
D. Pixelated tick, laterally reversed (diagram D).

Official answer: C

C. A coherent bundle preserves the relative positions of fibres, so it preserves image orientation. Each of the 36 fibres transmits one sampled image element, limiting resolution to a 6×6 array. The image is therefore pixelated without lateral reversal.

Question 4.3

Three musicians play notes P, Q and R of equal perceived loudness, as shown on the equal-loudness graph. A microphone picks up the performance and a loudspeaker reproduces each note at a sound intensity level 20 dB above its original level. What is the order of the reproduced loudness?



A. $P = Q = R$

B. $P > R > Q$

C. $R > Q > P$

D. $Q > R > P$

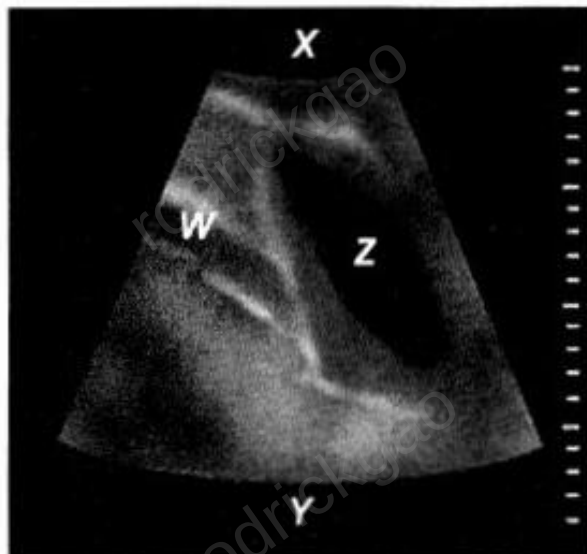
Official answer: B

B. At each frequency, move the original point vertically upward by 20 dB, then compare the equal-loudness contours reached. P reaches the highest contour, R the next and Q the lowest, giving $P > R > Q$. Equal changes in sound intensity level need not give equal changes in perceived loudness at different frequencies.

Question 4.4

The figure shows an ultrasound B-scan. Which statements are correct?

- (1) X is closer to the scanner than Y.
- (2) Z is dark because it absorbs more ultrasound.
- (3) W is bright because it reflects more ultrasound.



- A. (1) and (2) only
- B. (1) and (3) only**
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: B

B. The top of the sector is near the transducer, so X is nearer than Y. B-scan brightness represents returned echo strength: bright W gives strong echoes and dark Z gives weak echoes. Darkness alone does not show that the region absorbs more ultrasound.

Question 4.5

The intensity reflection coefficient between fat and muscle is 1%. Fat has acoustic impedance $1.38 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$. Estimate the muscle's acoustic impedance.

- A. $1.5 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$
- B. $1.6 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$
- C. $1.7 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$**
- D. $1.8 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$

Official answer: C

C. $R = [(Z_m - Z_f)/(Z_m + Z_f)]^2 = 0.01$, so the magnitude of the ratio is 0.1. All offered values exceed Z_f , giving $Z_m = (1.1/0.9)Z_f = 1.687 \times 10^6 \approx 1.7 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$. The squared equation alone also has a lower-impedance branch, absent from the options.

Question 4.6

A 5 cm thickness of tissue reduces an X-ray beam's intensity to 59% of its original value. What is the tissue's linear attenuation coefficient?

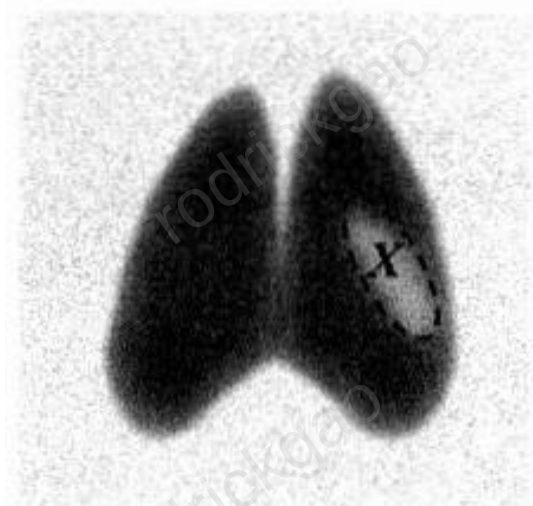
- A. 0.066 m^{-1}
- B. 0.085 m^{-1}
- C. 8.2 m^{-1}
- D. 10.6 m^{-1}

Official answer: D

D. $I/I_0 = \exp(-\mu x) = 0.59$. With $x = 0.050 \text{ m}$, $\mu = -\ln(0.59)/0.050 = 10.553 \approx 10.6 \text{ m}^{-1}$. The transmitted fraction is 0.59, not the lost fraction 0.41.

Question 4.7

The figure shows a thyroid scan using an iodine-131 tracer. Darker areas represent higher intensity detected by the gamma camera. Which deduction about region X is correct?



- A. It causes abnormally high attenuation of gamma radiation.
- B. It causes abnormally low attenuation of gamma radiation.
- C. It absorbs excessive iodine.
- D. It cannot absorb iodine normally.

Official answer: D

D. X is lighter than surrounding thyroid tissue, so it emits fewer detected gamma photons. In tracer imaging this indicates reduced uptake of iodine-131, a low-activity region. This is an emission image of tracer distribution, not a transmission image determined primarily by attenuation.

Question 4.8

Which statements explain why technetium-99m is suitable for medical radionuclide imaging?

- (1) It can combine with a wide range of chemicals and proteins to form radioactive tracers.
- (2) Its short half-life helps keep patient radiation exposure low.
- (3) Its gamma radiation is attenuated differently by tissues to give a radiographic image.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

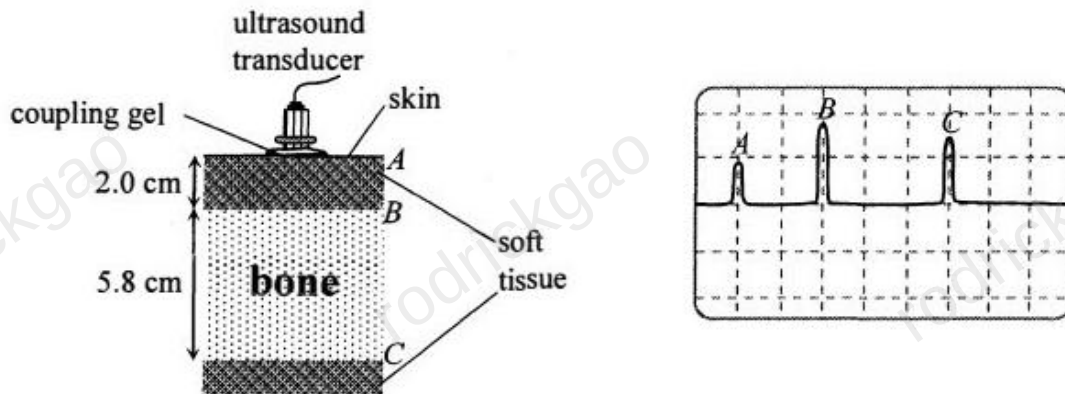
D. (1), (2) and (3)

Official answer: A

A. Chemical versatility permits tracers targeted to different organs, and a short half-life limits prolonged irradiation after the scan. The useful image records gamma emission from the tracer's distribution; differential attenuation of an external beam describes radiography, so (3) is not the correct imaging principle.

Question 4 Structured Question

(a) Figure 4.1 shows bone 5.8 cm thick below a 2.0 cm layer of soft tissue. An ultrasound transducer with coupling gel is applied to the skin. Echoes from boundaries A, B and C are shown on the CRO.



(i) Find the ratio of ultrasound speed in bone to that in soft tissue. (2 marks)

Acoustic impedances for the ultrasound used are:

Tissue	Acoustic impedance / $\text{kg m}^{-2} \text{s}^{-1}$
Soft tissue (average)	1.63×10^6
Bone	7.78×10^6

(ii) Ultrasound speed in soft tissue is 1580 m s^{-1} . Estimate the density of bone. (3 marks)

(b)(i) Describe the working principles of ultrasound B-scan imaging. (3 marks)

(ii) State ONE advantage and ONE limitation of ultrasound scans in medical imaging. (2 marks)

Official marking scheme

Marks

4. (a) (i) Let v be the speed of ultrasound in soft tissue
 v_b be the speed of ultrasound in bone
 t be the time of travel of ultrasound in soft tissue
 t_b be the time of travel of ultrasound in bone

$$\text{We have } \frac{v_b t_b / 2}{vt / 2} = \frac{5.8}{2.0} \text{ or } \frac{v_b t_b}{vt} = \frac{5.8}{2.0}$$

1M

$$\left(\frac{v_b}{v}\right)\left(\frac{3}{2}\right) = 2.9$$

$$\frac{v_b}{v} = 1.93$$

1A 2

- (ii) From (i), $v_b = 1.93 \times 1580 \text{ m s}^{-1} = 3055 \text{ m s}^{-1}$

1M

$$\therefore \text{ for bone } Z = \rho c$$

$$7.78 \times 10^6 = \rho (3055)$$

$$\rho = 2547 \text{ kg m}^{-3}$$

1M

1A 3

- (b) (i) Reflection of ultrasound at tissue boundary/when entering another tissue. 1A
 The brightness/amplitude/strength in a B-scan image is proportional to the intensity of the reflected ultrasound signal/change in acoustic impedance. 1A
 Distance/depth calculated from time for signal to return to ultrasound transducer./When traces are joined, a 2D/planar image of the scanned area is formed. 1A 3

- (ii) Advantages (ANY ONE):

1A

1. relatively safe (as no ionizing radiation involved)
2. readily accessible
3. can detect the movement of an organ in real time

Limitations (ANY ONE):

1A

1. limited tissue penetration, particularly through bones or air-filled structures
2. field-of-view (FOV) of ultrasound is smaller compared to other methods of imaging

2

(a)(i) A–B spans two time divisions and B–C three. Each interval is an extra round trip in one layer, so $t = 2(0.020)/v$ and $t_b = 2(0.058)/v_b$. Therefore $v_b/v = (0.058/0.020)(t/t_b) = 2.9 \times 2/3 = 1.9333 \approx 1.93$.

(a)(ii) Retain the unrounded ratio: $v_b = (29/15) \times 1580 = 3054.67 \approx 3055 \text{ m s}^{-1}$. Then $\rho = Z/v_b = 7.78 \times 10^6/3054.67 = 2546.9 \approx 2547 \text{ kg m}^{-3}$. The printed $1.93 \times 1580 = 3055$ equality uses a rounded displayed factor; the unrounded factor reproduces the official result.

(b)(i) The transducer sends short pulses and receives echoes from boundaries with different acoustic impedances. Echo delay gives depth through $d = vt/2$ in a uniform medium, and echo strength controls displayed brightness. Repeating along many lines builds a two-dimensional section. More precisely, at normal incidence the intensity reflection coefficient is $[(Z_2 - Z_1)/(Z_2 + Z_1)]^2$; it is not simply linear in impedance difference.

(b)(ii) One advantage is that ultrasound does not use ionizing radiation and can show motion in real time. One limitation is poor penetration through bone or air because strong boundary reflections leave little transmitted signal for deeper structures.